

Tarea Sección 7.4

$$7) \int \frac{x}{x-6} dx = \int \frac{1}{x-6} = \int \frac{1}{m} dm = \ln|x-6|$$

$$m = x-6$$

$$dm = dx$$

$$9) \int \frac{x-9}{(x+5)(x-2)} dx = \int \frac{A}{x+5} + \frac{B}{x-2} = \int \frac{2}{x+5} + \int \frac{-1}{x-2} = 2 \int \frac{1}{m} dm - \int \frac{1}{n} dn$$

$$\frac{A}{x+5} + \frac{B}{x-2} = \frac{A(x-2) + B(x+5)}{(x+5)(x-2)} = \frac{Ax - 2A + Bx + 5B}{(x+5)(x-2)} = \frac{(A+B)x - 2A + 5B}{(x+5)(x-2)}$$

$$\begin{aligned} A+B &= 1 & A &= 2 \\ -2A+5B &= 9 & B &= -1 \end{aligned} \quad \begin{aligned} m &= x+5 & n &= x-2 \\ dm &= dx & dn &= dx \end{aligned} \quad 2 \ln|x+5| - \ln|x-2| + C$$

$$11) \int_0^1 \frac{2}{2x^2+3x+1} dx = \int_0^1 \frac{2}{(x+1)(2x+1)} dx = \int \frac{A}{x+1} + \frac{B}{2x+1} = -\int \frac{2}{x+1} + \int \frac{4}{2x+1} =$$

$$\frac{A}{x+1} + \frac{B}{2x+1} = \frac{A(2x+1) + B(x+1)}{(x+1)(2x+1)} = \frac{2Ax + A + Bx + B}{(x+1)(2x+1)} = \frac{(2A+B)x + A+B}{(x+1)(2x+1)}$$

$$\begin{aligned} 2A+B &= 0 & 2(2-B)+B &= 0 & -2 \int \frac{1}{m} + 4 \int \frac{1}{n} \frac{dn}{2} \\ A+B &= 2 & 4-2B+B &= 0 \\ A &= 2-B & 4-B &= 0 & -2 \ln|x+1| + 2 \ln|2x+1| \Big|_0^1 \approx 0.87 \\ & & B &= 4 \end{aligned}$$

$$A+4=2$$

$$A=-2$$

$$m = x+1$$

$$dm = dx$$

$$n = 2x+1$$

$$dn = 2dx$$

$$13) \int_3^4 \frac{x^3 - 2x^2 - 4}{x^3 - 2x^2} dx = \int_3^4 1 - \frac{4}{x^3 - 2x^2} dx = \int_3^4 1 - \frac{4}{x^2(x-2)} = \int_3^4 1 dx - \int_3^4 \frac{4}{x^2(x-2)} dx$$

$$\frac{x^3 - 2x^2}{x^3 - 2x^2} \frac{4}{x^2(x-2)} = \frac{4}{x^2(x-2)}$$

$$= \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-2} = \frac{Ax(x-2) + B(x-2) + Cx^2}{x^2(x-2)}$$

$$Ax^2 - 2Ax + Bx - 2B + Cx^2 = (A+C)x^2 + (-2A+B)x - 2B$$

$$A+C=0 \quad -1+C=0$$

$$-2A+B=0 \quad -2A-2=0$$

$$-2B=4 \quad B=-2$$

$$C=1 \quad A=-1$$

$$= \int \frac{1}{x} + 2 \int \frac{1}{x^2} + \int \frac{1}{x-2} = \frac{1}{m} dm$$

$$= x + \ln|x| - 2\frac{1}{x} - (\ln|x-2|) \Big|_3^4$$

$$[4.19 - 3.43] = 0.76$$

$$15) \int_{-1}^0 \frac{x^3 - 4x + 7}{x^2 - 3x + 2} dx = \int_{-1}^0 x dx + \int_{-1}^0 3 dx + \int_{-1}^0 \frac{3x-5}{x^2-3x+2} dx =$$

$$\int \frac{3x-5}{(x-1)(x-2)} dx = \int \frac{A}{x-1} dx + \int \frac{B}{x-2} dx$$

$$A(x-2) + B(x-1) = Ax - 2A + Bx - B$$

$$(A+B)x - 2A - B$$

$$A=3-1$$

$$A=2$$

$$A+B=3$$

$$-2A-B=-5$$

$$-2(3-B)-B=-5$$

$$-6+2B-B=-5$$

$$B=1$$

$$\int_{-1}^0 x dx + \int_{-1}^0 3 dx + \int_{-1}^0 \frac{2}{x-1} dx + \int_{-1}^0 \frac{1}{x-2} dx$$

$$\frac{x^2}{2} \Big|_{-1}^0 + 3x \Big|_{-1}^0 + 2 \int_{-1}^0 \frac{1}{m} dm + \int_{-1}^0 \frac{1}{n} dn$$

$$\frac{1}{2} - 3 + 2(\ln|x-1|) + (\ln|x-2|)$$

$$0.0931471806 - (-0.01509335) = 0.70$$

$$77) \int_1^2 \frac{4y^2 - 7y - 12}{y(y+2)(y-3)} dy = \int \frac{A}{y} + \int \frac{B}{y+2} + \int \frac{C}{y-3} =$$

$$\frac{A(y+2)(y-3) + B(y-3) + C(y)(y+2)}{y(y+2)(y-3)} = \frac{Ay^2 - 3Ay + 2Ay - 6A + By^2 - 3By + Cy^2 + 2Cy}{y(y+2)(y-3)}$$

$$(A+B+C)y^2 + (-3A+2A+2C-3B)y - 6A = (A+B+C)y^2 + (-A+2C-3B)y - 6A$$

$$A+B+C=4 \quad 2+B+C=4 \quad B=2-C \quad B=\frac{9}{5}$$

$$-A+2C-3B=-7 \quad -2+2C-3(2-C)=-7 \quad 2C-6+3C=-5$$

$$-6A=-12$$

$$A=2$$

$$5C=1$$

$$C=\frac{1}{5}$$

$$\int \frac{2}{y} + \int \frac{9}{y+2} + \int \frac{1}{y-3} = 2 \int \frac{1}{y} + \frac{9}{5} \int \frac{1}{y+2} + \frac{1}{5} \int \frac{1}{y-3}$$

$$2(\ln|y|) + \frac{9}{5}(\ln|y+2|) + \frac{1}{5}(\ln|y-3|) \Big|_1^2 = 3.881024 - 2.110131 = 1.76$$

$$79) \int_0^1 \frac{x^2+x+1}{(x+1)^2(x+2)} dx = \int \frac{A}{x+1} dx + \int \frac{B}{(x+1)^2} dx + \int \frac{C}{x+2} dx = \frac{A(x+1)(x+2) + B(x+1) + C(x+1)^2}{(x+1)^2(x+2)}$$

$$Ax^2 + 2Ax + Ax + 2A + Bx + 2B + Cx^2 + 2Cx + C =$$

$$(A+C)x^2 + (2A+A+B+2C)x + 2A+2B+C$$

$$(A+C)x^2 + (3A+B+2C)x + 2A+2B+C =$$

$$-2 \int \frac{1}{x+1} dx + \int \frac{1}{(x+1)^2} dx + 3 \int \frac{1}{x+2} dx$$

$$A+C=1 \quad A=1-C$$

$$3A+B+2C=1$$

$$2A+2B+C=1$$

$$A=-2, B=1, C=$$

$$m=x+1$$

$$dm=dx$$

$$-2(\ln|x+1| - \frac{1}{x+1}) + 3(\ln|x+2|) \Big|_0^1$$

$$1.4095 - 1.0794 \approx 0.3301$$

$$\int \frac{1}{m^2} dm$$

$$m^{-2}$$

$$m^{-1}$$

$$-1$$

$$27) \int \frac{dt}{(t^2-1)^2} = \int \frac{dt}{(t^2-1)^2 (t^2-1)^2}$$

$$= \int \frac{A}{t-1} + \int \frac{B}{(t-1)^2} + \int \frac{C}{t+1} + \int \frac{D}{(t+1)^2} = \frac{A}{(t^2-1)^2 (t^2-1)^2}$$

$$= \int \frac{dt}{[(t+1)(t-1)]^2} = \int \frac{dt}{(t+1)^2 (t-1)^2}$$

$$= \int \frac{A}{t+1} dt + \int \frac{B}{(t+1)^2} dt + \int \frac{C}{t-1} dt + \int \frac{D}{(t-1)^2} dt$$

$$\frac{A(t+1)(t-1)^2 + B(t-1)^2 + C(t+1)^2(t-1) + D(t+1)^2}{(t+1)^2(t-1)^2}$$

$$At^3 - 2At^2 + At + At^3 - 2At + A + Bt^2 - 2Bt + B + Ct^3 + 2Ct + Ct - Ct^2 - 2Ct - C + Dt^2 + 2Dt + D$$

$$(A+C)t^3 + (-2A+A+B+2C-C+D)t^2 + (A-2A-2B+C-2C+2D)t + A+B-C+D$$

$$A+C=0 \quad A=\frac{1}{4} \quad = \int \frac{\frac{1}{4}}{t+1} dt + \int \frac{\frac{1}{4}}{(t+1)^2} dt + \int \frac{-\frac{1}{4}}{t-1} dt + \int \frac{\frac{1}{4}}{(t-1)^2} dt$$

$$-A+B+C+D=0 \quad B=\frac{1}{4}$$

$$-A-2B-C+2D=0 \quad C=-\frac{1}{4} \quad = \frac{1}{4} \int \frac{1}{m} dm + \frac{1}{4} \int \frac{1}{n^2} - \frac{1}{4} \int \frac{1}{m} dm + \frac{1}{4} \int \frac{1}{n^2} dt$$

$$A+B-C+D=1 \quad D=\frac{1}{4}$$

$$= \frac{1}{4} (n|t+1|) + \frac{1}{4} \left(-\frac{1}{t+1}\right) - \frac{1}{4} (n|t-1|) + \frac{1}{4} \left(-\frac{1}{t-1}\right) + C$$

$$= \frac{1}{4} (n|t+1|) - \frac{1}{4(t+1)} - \frac{1}{4} (n|t-1|) - \frac{1}{4(t-1)}$$

$$n=t+1$$

$$dn=dt$$

$$23) \int \frac{10}{(x-1)(x^2+9)} = \int \frac{A}{x-1} + \int \frac{Bx+C}{x^2+9}$$

$$\frac{A(x^2+9) + Bx(x-1) + C(x-1)}{(x-1)(x^2+9)} = \frac{Ax^2 + 9A + Bx^2 - Bx + Cx - C}{(x-1)(x^2+9)} = \frac{(A+B)x^2 + (-B+C)x + 9A - C}{(x-1)(x^2+9)}$$

$$A+B=0$$

$$-B+C=0$$

$$9A-C=10$$

$$A=-B$$

$$-1=-B \quad B=-1$$

$$A+9A-10=0$$

$$9A-10=C$$

$$C=-1$$

$$B=-1$$

$$10A-10=0$$

$$A=1$$

$$\int \frac{1}{x-1} + \int \frac{-1x}{x^2+9} + \int \frac{-1}{x^2+9} = \int \frac{1}{m} dm - \int \frac{x}{n} \frac{dn}{2x} - \int \frac{1}{n} dn$$

$$- \int \frac{1}{x^2+9} dx$$

$$m=x-1$$

$$n=x^2+9$$

$$dm=dx$$

$$dn=2x dx$$

$$K = \frac{x}{3} \Rightarrow$$

$$3K=x$$

$$3dK = \frac{1}{3} dx \Rightarrow 3dK = dx$$

$$\int \frac{1}{9x^2+9}$$

$$3dK \Rightarrow$$

$$\int \frac{1}{9(K^2+1)}$$

$$\frac{1}{3} \int \frac{1}{K^2+1}$$

$$= \ln|x-1| - \frac{1}{2} \ln|x^2+9| - \frac{1}{3} \tan^{-1}\left(\frac{x}{3}\right) + C$$

$$25) \int \frac{4x}{x^3+x^2+x+1} dx = \int \frac{4x}{(x+1)(x^2+1)} dx = \int \frac{A}{x+1} dx + \int \frac{Bx+C}{x^2+1}$$

$$\frac{(x^2+x^2)(x+1)}{(x+1)(x^2+1)} \quad A(x+1) + Bx(x+1) + C(x+1) = Ax^2 + A + Bx^2 + Bx + Cx + C$$

$$(x+1)(x^2+1) \quad Bx^2 + (A+B+C)x + A+C$$

$$Ax^2 + A + Bx^2 + Bx + Cx + C$$

$$A+B=0 \quad A+(4-C)=0$$

$$B+C=4 \quad B=4-C$$

$$A+C=0 \quad C=-A$$

$$(A+B)x^2 + (B+C)x + A+C$$

$$= \int \frac{-2}{x+1} dx + \int \frac{2x}{x^2+1} dx + \int \frac{2}{x^2+1} dx$$

$$= -2 \ln|x+1| + \ln|x^2+1| + 2 \tan^{-1}x + C$$

$$m = x^2$$

$$dm = 2x dx$$

$$A+4+A=0$$

$$2A = -4$$

$$A = -2$$

$$C = 2$$

$$B = 4 - 2 = 2$$

$$\frac{1}{4} \frac{x}{m} \frac{dm}{2x}$$

$$27) \int \frac{x^3+4x+3}{x^4+5x^2+4} dx = \int \frac{x^3+4x+3}{(x^2+4)(x^2+1)} dx = \int \frac{Ax+B}{x^2+4} dx + \int \frac{Cx+D}{x^2+1}$$

$$x^4+4x^2+x+4$$

$$= \frac{Ax(x^2+1) + B(x^2+1) + Cx(x^2+4) + D(x^2+4)}{(x^2+4)(x^2+1)}$$

$$(x^4+4x^2)(x^2+1) =$$

$$\frac{x^2(x^2+4)(x^2+1)}{(x^2+4)(x^2+1)}$$

$$Ax^3 + Ax + Bx^2 + B + Cx^3 + 4Cx + Dx^2 + 4D$$

$$m = x^2+4$$

$$dm = 2x dx$$

$$\frac{x}{m} \frac{dm}{2x}$$

$$(A+C)x^3 + (B+D)x^2 + (A+4C)x + B+4D$$

$$A+C=1 \quad A=1-C$$

$$B+D=0 \quad B=-D \quad B=-1$$

$$A+4C=4 \quad 1-C+4C=4$$

$$B+4D=3$$

$$3C=3$$

$$C=1$$

$$A=0$$

$$-D+4D=3$$

$$D=1$$

$$= \int \frac{0x}{x^2+4} + \int \frac{-1}{x^2+4} + \int \frac{1x}{x^2+1} + \int \frac{1}{x^2+1}$$

$$= -\frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) + \frac{1}{2} \ln|x^2+1| + \tan^{-1}(x)$$

$$29) \int \frac{x+4}{x^2+2x+5} dx = \int \frac{\overset{m+3}{(x+1)+3}}{(x+1)^2+4} dx = \int \frac{m+3}{m^2+4} = \int \frac{m}{m^2+4} + \int \frac{3}{m^2+4} dm$$

$$(x+1)^2 - 1 + 5 = \int \frac{m}{n} \frac{dn}{2m} + 3 \int \frac{1}{m^2+4} dm$$

$$= \frac{1}{2} \int \frac{1}{n} dn$$

$$= \frac{1}{2} (\ln |(x+1)^2+4| + \frac{3}{2} \tan^{-1}(\frac{m}{2}))$$

$$= \frac{1}{2} (\ln |x^2+2x+5| + \frac{3}{2} \tan^{-1}(\frac{x+1}{2})) + C$$

$$m = x+1$$

$$n = m^2+4$$

$$dn = 2m dm$$